

Methodology

STAGE 1 · BEHAVIORAL

Run prompt variants, measure wrong-answer rate

Three variants per question, same decoding settings.

N0 no cue

C1 correct endorsement

W1 wrong endorsement

STAGE 2 · EXTRACT $\hat{d}_{\text{authority}}$

Fit the authority direction

Cache activations from N0 / C1 / W1 prompts at every (layer, position).

Pick the slot with the cleanest held-out AUROC.

$$\hat{d}_{\text{authority}} = \text{unit}(\mu_{\text{W1-complying}} - \mu_{\text{C1-complying}})$$

STAGE 3 · PROBE

Forward-patching

Add $\hat{d}_{\text{authority}}$ to a clean (N0) prompt; check whether the model behaves as if cued.

$$h' = h + \alpha \cdot \hat{d}_{\text{authority}}$$

↑ matched flip rate (right → wrong)

STAGE 4 · REMOVE

Projection-removal

Subtract a chosen direction $\hat{v}$ from a wrong-cue (W1) prompt; surgically zero that component.

$$h' = h - \alpha(h \cdot \hat{v})\hat{v}$$

↓ wrong-rate reduction (wrong → right)

$\hat{v} \in \{ \hat{d}_{\text{authority}}, \hat{d}_{\text{assistant}}, \hat{d}_{\text{authority}-\text{assistant}}, \hat{d}_{\text{source}}, \hat{d}_{\text{user}} \}$

Projection removal

